

**Report
Of
Java-Based Student Management System**

Submitted

To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

**In partial fulfilment of the requirement of degree of
B.Tech CSE**



By

Roll Number of Student: 25SCS1003005290

Name of Student: ANKIT

Batch: 2026-27

CANDIDATE'S DECLARATION

I, ANKIT, student of B.Tech CSE, 3rd Semester, Section 2CSE27, Roll No. 25SCS1003005290, IILM University, Greater Noida, hereby declare that this report entitled “Java-Based Student Management System” has been prepared as part of my internship documentation for the Java Developer Internship completed at Codec Technologies Pvt. Ltd.

The report presents the learning and project-based application of Core Java concepts including object-oriented programming, classes, objects, methods, collections, exception handling and software development practices.

I declare that this report has been prepared for academic submission and that the sources used for technical concepts have been acknowledged in the Bibliography/References section.

Name of Student: ANKIT

Roll Number: 25SCS1003005290

Signature of Student

ACKNOWLEDGMENT

I would like to express my sincere gratitude to IILM University, Greater Noida, for providing me with the opportunity to undertake this internship and prepare the report.

I am sincerely thankful to my faculty guide, Mr. Anik, for his guidance and encouragement. I also express my gratitude to Codec Technologies Pvt. Ltd. for the internship exposure in the role of Java Developer Intern.

The internship helped me strengthen my understanding of Core Java, object-oriented programming, arrays, methods, classes, objects, collections, exception handling and basic project development.

ANKIT

Roll No. 25SCS1003005290

INTERNSHIP COMPLETION CERTIFICATE



TABLE OF CONTENTS

Chapter No.	Chapter Name	Page Nos.
	Cover Page	----
1.	Candidate's Declaration	
2.	Acknowledgment	
3.	Internship Completion Certificate	
4.	Project Description 4.1 Introduction 4.2 Organization Profile 4.3 Problem Statement 4.4 Project Objectives 4.5 Scope of the Project 4.6 Technologies and Tools Used 4.7 System Architecture 4.8 Methodology 4.9 Expected Outcomes 4.10 Certificates of completion and other communication mail proof must be attached	
5.	Bibliography/References	

Instruction for the report format

- Cover Page, Copy of the Certificate issued by industry/organisation will have no Page Number.
- Page no. of Sr. No. 1 to 6 along with a list of tables and list of figures will bear the Roman Numbering System (e.g. i, ii, iii).
- Bibliography/References should be in proper format.

Size of Paper:	A4
Printing:	Single Sided
Left Margin:	1.5 inch
Right Margin:	1 inch
Top Margin:	0.75 inch
Bottom Margin:	1 inch (with Page No. at the centre)

Page Numbering:	Centered at the bottom of the Page
Line Spacing:	1.5 for text and 1 for References
Paragraph:	Justified
Font Type:	Times New Roman
Font Size:	12 for text and within tables can vary from 8-12 (But should be same throughout the report)
Heading Font Size:	12 for Sub Headings and 14 for main Headings with Bold Text
Cover Page:	As Specified Above
Tables, Figures and Graphs:	With Numbers indicating the chapters (1.1, 1.2 or 2.1, 2.2 etc.) and contain titles (above for tables)

LIST OF TABLES

Table 4.1 Functional Requirements

Table 4.2 Technologies and Tools Used

Table 4.3 Module-wise Project Description

Table 4.4 Test Cases and Results

LIST OF FIGURES

Figure 4.1 System Architecture

Figure 4.2 Application Workflow

Figure 4.3 Module Interaction Overview

CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

The Java Developer Internship focused on understanding and applying Core Java concepts through a practical software project. The selected project is a Java-Based Student Management System designed to manage student records through adding, viewing, searching, updating and deleting information.

The project demonstrates how classes and objects, methods, conditional statements, loops, collections and exception handling can be combined into a structured application. It is suitable for demonstrating Core Java concepts in a practical and understandable way.

4.2 Organization Profile

Codec Technologies Pvt. Ltd. is the organization mentioned on the internship completion certificate. The internship was completed in the role of Java Developer Intern from 02/07/2026 to 02/08/2026.

The internship learning documented in this report focuses on Core Java concepts such as object-oriented programming, classes, objects, methods, arrays, collections, exception handling and basic software design practices.

4.3 Problem Statement

Managing student information manually becomes difficult when records increase. Searching, updating and deleting records can require repeated effort and may lead to errors.

The problem addressed by the project is to develop a simple Core Java application that manages student records through a menu-driven interface and demonstrates practical object-oriented programming.

4.4 Project Objectives

- To strengthen practical understanding of Core Java.
- To apply object-oriented programming concepts in a mini project.
- To create a simple student record management application.
- To implement add, view, search, update and delete operations.

- To use Java collections and exception handling.
- To improve problem-solving, debugging and testing skills.

Table 4.1 Functional Requirements

Requirement	Description
Add Student	Stores a new student record.
View Students	Displays all student records.
Search Student	Finds a student by roll number.
Update Student	Modifies selected information.
Delete Student	Removes a student record.

4.5 Scope of the Project

The project scope is limited to a Core Java console-based application for managing basic student records. It is intended for academic learning and demonstration rather than production deployment.

- Menu-driven console interface.
- Student record creation and display.
- Search, update and delete operations.
- Basic input validation.
- Use of classes, objects, methods and collections.
- Modular structure for future enhancement.

4.6 Technologies and Tools Used

Core Java is used as the primary programming technology. A Java Development Kit is required to compile and execute the program, while an IDE or code editor can be used for development and debugging.

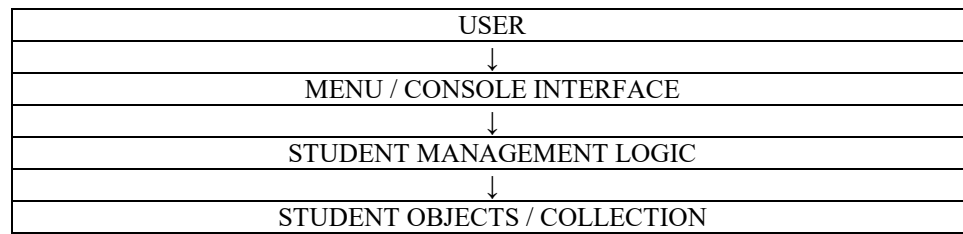
Table 4.2 Technologies and Tools Used

Technology / Tool	Purpose
Core Java	Application logic and OOP.
JDK	Compiling and executing Java.
IDE / Code Editor	Writing and debugging code.
Java Collections	Managing multiple records.
Exception Handling	Handling invalid input and errors.

4.7 System Architecture

The system follows a simple logical architecture: User → Menu Interface → Student Management Logic → Student Objects/Collection. This separation makes the program easier to understand and modify.

Figure 4.1 System Architecture

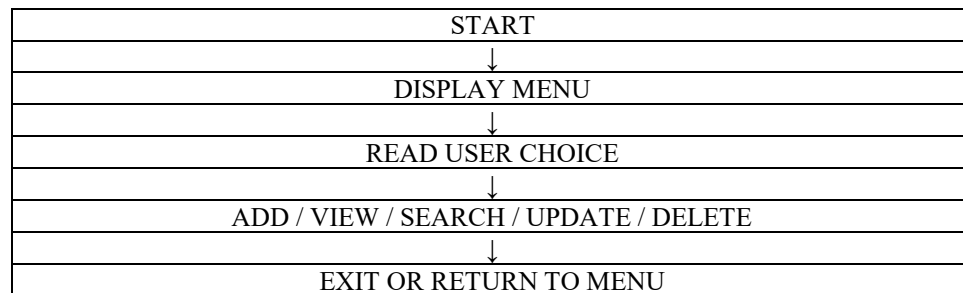


4.8 Methodology

The project methodology follows requirement identification, system design, implementation, validation, testing, debugging and documentation.

- Requirement Identification
- System Design
- Implementation
- Validation
- Testing
- Debugging
- Documentation

Figure 4.2 Application Workflow



4.9 Expected Outcomes

The expected outcome is improved practical understanding of Core Java and greater confidence in applying classes, objects, methods, collections and exception handling to a complete mini project.

Table 4.3 Module-wise Project Description

Module	Main Work
Student Model	Defines student attributes and methods.
Add Module	Creates and stores student records.
View Module	Displays available records.
Search Module	Finds a student by roll number.
Update Module	Modifies student information.
Delete Module	Removes a selected record.

4.10 Certificates of Completion and Other Communication

The internship completion certificate issued by Codec Technologies Pvt. Ltd. is attached as supporting evidence. Any additional official completion-related communication or email proof required by the University may be attached in this section.

Project Testing

Table 4.4 Test Cases and Results

Test Case	Input / Action	Expected Result
TC01	Add valid student	Student added.
TC02	View records	Records displayed.
TC03	Search existing roll number	Record shown.
TC04	Search unknown roll number	Not found message.
TC05	Update record	Record modified.
TC06	Delete record	Record removed.
TC07	Invalid menu choice	Validation message.

CHAPTER 5

BIBLIOGRAPHY / REFERENCES

1. Herbert Schildt, Java: The Complete Reference, McGraw-Hill Education.
2. Cay S. Horstmann, Core Java: Fundamentals, Pearson.
3. Oracle Java Documentation, Java Language and API Reference.
4. Oracle Java Documentation, Object-Oriented Programming, Collections and Exception Handling.
5. Codec Technologies Pvt. Ltd., Internship Completion Certificate.